

The Contaminated City

AP Chemistry → college analytical

A wide river city, three days after a freight-yard fire.

45	10	6.7	7/12	2	13 · 31 · 1
STOPS	CHOICE (22%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Curriculum is in reasonable shape — 7 of 12 equations computed and only three select-only concepts — and the mix is already the fourth-best in the set. Three of the five gaps sit on stops that look like they compute the thing (a BALLPARK that predicts a freezing point, a BALLPARK that audits charge passed), so Phase 0 is likely to clear them. The one that will survive is the solubility product, which lives on a CHOICE and nowhere else.

NOW The mix as it ships

CHOICE 10	BALLPARK 9	PROTOCOL 8	SEQUENCE 6	DIAGNOSIS 3	VALUE 2	TRACE 1	ALLOCATE 1	VERIFY 1	TRIGGER 1	STRESS 1	BALANCE 1	CLOUD 1
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CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$\Delta T_f = i \cdot K_f \cdot m$	how far dissolved salt drops a freezing point	24 · BALLPARK	real gap
$n = It \div (zF)$	how much a current can deposit or remove	31 · PROTOCOL 44 · BALLPARK	real gap
$A = \epsilon lc$	Beer–Lambert, and the calibration curve it justifies	17 · VERIFY	verify first
$K_{sp} = [A^a] [B^b]$	the solubility product and the common-ion effect	14 · CHOICE	real gap
$rate = k[A]^n$	a rate law, and the order it asserts	9 · BALLPARK 34 · PROTOCOL 35 · SEQUENCE 36 · STRESS	verify first

2 of 5 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 3 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
25	Solubility product and the common-ion effect	14
27	Mass spectrometry and molecular mass	40

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 1 concept no stop touches at all — **Colligative properties** — which is expected, and is a different conversation from this one.

FUN What this subject and this place can carry

PROBE STRONG

a new WATER or GASES stop

The place is a river city with an intake laboratory at one end and a plume station at the other. PROBE hands over no readings: you sample station by station down the river and name where the pattern changes — which is not the station with the worst number on it.

SERVES transport and dilution; the plume as a gradient rather than a value

NEEDS a chain of stations with authored readings and one honest break

SWEEP STRONG

stop 22 or 24 · ENERGY

A rate law is a slope, and it is the one thing on this syllabus that nothing computes. Drag concentration across its range and the response is plotted only where you look; the order is the shape you built.

SERVES $\text{rate} = k[A]^n$ — the gap with the most mentions in the game

NEEDS a plateau that beats a spike, and `showControls()` so the sticky row does not eat the slider

HOLD STRONG

TREAT · the pilot plant

Hold pH inside a closing band while a load of acid arrives. A buffer is exactly a quantity that resists being pushed until it does not, and the band closing as capacity is consumed is the lesson.

SERVES buffers and titration, currently taught only as prose

NEEDS the band's closing rate authored from buffer capacity, not chosen for difficulty

SPOT MEDIUM

QUANT · sample intake

A QA acceptance criterion replaced mid-run with nothing announcing it. Real, and it is what a quality office does. Weaker here than elsewhere only because the syllabus item it carries is method rather than chemistry.

SERVES chain of custody and acceptance criteria

RISK score only the discriminating items, or a run that ignores the change passes

BELT MEDIUM

RECORDS or QUANT

A receiving bench sorting arrivals into destructive and non-destructive workflows. The game already has an ordering stop about exactly that axis, so the one bit of subject matter is available and pre-argued.

SERVES analytical workflow order

RISK must be decidable at a glance or it becomes dexterity

TRIAL MEDIUM

the city

Sampling order across a city — upstream before downstream, air before water — driven rather than sequenced. It works, but Riverton is wide and the drive is long for one stop.

SERVES sampling design

RISK the most expensive item here; gates must stand off each door or they render under the floor

LOB REJECT

Nothing in analytical chemistry is aimed. Rejected on subject, not on taste.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: `curriculumDelivery.mjs contamcity --snapshot before.json`, then `--against before.json`.

Minimum honest 3 STOPS · ONE SITTING

14	d5	WATER	CHOICE → CLOUD	Use the common-ion effect to clear the limit the solubility product against a limit, where the common ion narrows nothing
34	d12	ENERGY	PROTOCOL → SWEEP	What changes the rate? the rate law as a slope you build by looking
10	d4	IDENT	CHOICE → TRIGGER	What the position in the table tells you periodic trends as a rule written before the element arrives

CHOICE **10 → 8** (18%) · moves **13·31·1 → 11·32·2** · effective formats **6.7 → 7.9** · **1** new operate stop

Balanced 6 STOPS · THE RECOMMENDATION

14	d5	WATER	CHOICE → CLOUD	Use the common-ion effect to clear the limit
34	d12	ENERGY	PROTOCOL → SWEEP	What changes the rate?
10	d4	IDENT	CHOICE → TRIGGER	What the position in the table tells you
40	d14	QUANT	CHOICE → PROBE	Reading a mass spectrum a mass spectrum read peak by peak rather than picked from four
31	d11	TREAT	PROTOCOL → CHAIN	Find the anode and cathode the electrode path, and the step that governs the deposit
36	d12	ENERGY	STRESS → HOLD	Define the safe operating envelope the safe envelope held rather than defined

CHOICE **10 → 7** (16%) · moves **13·31·1 → 10·31·4** · effective formats **6.7 → 8.8** · **3** new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

14	d5	WATER	CHOICE → CLOUD	Use the common-ion effect to clear the limit
34	d12	ENERGY	PROTOCOL → SWEEP	What changes the rate?
10	d4	IDENT	CHOICE → TRIGGER	What the position in the table tells you
40	d14	QUANT	CHOICE → PROBE	Reading a mass spectrum
31	d11	TREAT	PROTOCOL → CHAIN	Find the anode and cathode
36	d12	ENERGY	STRESS → HOLD	Define the safe operating envelope
35	d12	ENERGY	SEQUENCE → SPOT	Build a runaway feedback loop a runaway whose acceptance criterion changes under you
17	d6	QUANT	VERIFY → TALLY	Protect the calibration calibration points accumulated until the curve is defensible
9	d3	ENERGY	BALLPARK → BELT	Heat release is not the spontaneity test exotherm or not, sorted at the speed a receiving bench works at

CHOICE **10 → 7** (16%) · moves **13·31·1 → 10·29·6** · effective formats **6.7 → 9.9** · **6** new operate stops, 3 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — tools/import-book.mjs books/contamcity.yml contamcity --verify. **4.** npm run traps: every instrument's trap must fire. **5.** npm run check contamcity and npm run drive contamcity, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/contamcity/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=contamcity npm run dev`.